Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Let L_1 be the line $\vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda(\hat{i} + 2\hat{k})$ and let L_2 be the line $\vec{r}_2 = 3\hat{i} + \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$
Let Π be the plane which contains the line L_1 and is parallel to L_2. The distance of the plane Π from the origin is

(A) $1/7$

(B) $\sqrt{2/7}$

(C) $\sqrt{6}$

(D) None

Q.2

The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$ where $a, b \in N$ and $a > 3$ then $(a + b)$ equals

(A) 23

(B) 95

(C) 27

(D) None

Q.3

The Cartesian equation of a line is $\frac{x+3}{2} = \frac{y-5}{4} = \frac{z+6}{2}$. Find the vector equation for the line.

(A) $\overrightarrow{r} = (-3\hat{i} + 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} + 4\hat{j} + 2\hat{k})$

(B) $\overrightarrow{r} = (3\hat{i} - 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} + 4\hat{j} + 2\hat{k})$

(C) $\overrightarrow{r} = (3\hat{i} - 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} - 4\hat{j} + \hat{k})$

(D) $\overrightarrow{r} = (3\hat{i} + 5\hat{j} - 6\hat{k}) + \lambda(2\hat{i} - 2\hat{j} - \hat{k})$

Q.4

The distance between the line $x = 2 + t, y = 1 + t, z = \frac{-1}{2} - \frac{t}{2}$ and the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} + 6\hat{k}) = 10$, is

(A) $\frac{1}{6}$

(B) $\frac{1}{\sqrt{41}}$

(C) $\frac{1}{7}$

(D) $\frac{9}{\sqrt{41}}$

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Q.5

The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ will be perpendicular, if and only if

(A) $aa' + bb' + cc' + 1 = 0$

(B) $aa' + bb' + cc' = 0$

(C) $(a + a')(b + b') + (c + c') = 0$

(D) $aa' + cc' + 1 = 0$

Q.6

Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$.

(A) $\sin^{-1}\left(\frac{8}{21}\right)$

(B) $\sin^{-1}\left(\frac{5}{21}\right)$

(C) $\sin^{-1}\left(\frac{7}{21}\right)$

(D) $\sin^{-1}\left(\frac{1}{21}\right)$

Q.7

The position vector of a point in which a line through the origin perpendicular to the plane $2x - y - z = 4$ meets the plane $\vec{r} \cdot (3\hat{i} - 5\hat{j} + 2\hat{k}) = 6$, is

(A) $(1, -1, -1)$

(B) $(-1, -1, 2)$

(C) $(4, 2, 2)$

(D) $\left(\frac{4}{3}, \frac{-2}{3}, \frac{-2}{3}\right)$

Q.8

Let $A(-1, 0, 0), B(0, 1, 0), C(0, 0, -1)$ are the vertices of $\triangle ABC$. If R and r denote the circumradius and inradius of $\triangle ABC$, then the value of $\left(R^2 + r^2\right)$ is equal to

(A) $\frac{5}{6}$

(B) $\frac{1}{6}$

(C) $\frac{2}{3}$

(D) $\frac{1}{3}$

Q.9

A and B are points with coordinates $(2, 1, 3)$ and $(4, -2, 5)$ respectively $2\mathrm{x} + \mathrm{y} - \mathrm{z} = 3$ is the equation of a plane -

(A) Line AB is parallel to the plane

(B) AB meets the plane at $(0, 4, 1)$

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(C) The plane divides AB externally in the ratio $1:2$

(D) A and B are on the same side of the plane

Q.10

Which of the following statement(s) is/are correct :

(A) If $\vec{a},\vec{b},\vec{c}$ are non-coplanar and $\vec{d}$ is any vector, then $[\vec{d}\vec{b}\vec{c}]\vec{a}+[\vec{d}\vec{c}\vec{a}]\vec{b}+[\vec{d}\vec{a}\vec{b}]\vec{c}-[\vec{a}\vec{b}\vec{c}]\vec{d}=\overrightarrow{0}$

(B) If I is incentre of $\triangle ABC$ then $|\overrightarrow{\mathrm{BC}}|\overrightarrow{\mathrm{IA}}+|\overrightarrow{\mathrm{CA}}|\overrightarrow{\mathrm{IB}}+|\overrightarrow{\mathrm{AB}}|\overrightarrow{\mathrm{IC}}=\overrightarrow{0}$

(C) Any vector in three dimension can be written as linear combination of three non-coplanar vectors.

(D) In a triangle, if position vector of vertices are $\vec{a},\vec{b},\vec{c}$, then position vector of incentre is $\frac{\vec{a}+\vec{b}+\vec{c}}{3}$

Q.11

Find the values less than the square of the distance between the origin and the image of the point $(1,2,3)$ in the line $\frac{x}{-2}=\frac{y+1}{2}=\frac{z-1}{1}$ is $-$

(A) $\frac{46}{3}$

(B) 15

(C) $\frac{23}{3}$

(D) $\frac{35}{3}$

Q.12

Passage:
Let L be the line whose equations are $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{1}$. Let P represent the point $(1,2,3).L$ meets the plane π given by the equation $x+y+z=10$ at A. Produce PA to B such that PA $=$ AB. Let C be the image of B in π. Let D be a point on the plane π such that $ABDC$ is a parallelogram. Let L_1 be the projection of L on π.
Question:
Co-ordinates of B are -

(A) $(3,-6,5)$

(B) $(-3,6,5)$

(C) $(3,6,5)$

(D) $(-3,-6,5)$

Q.13

Passage:
Let L be the line whose equations are $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{1}$. Let P represent the point $(1,2,3).L$ meets the plane π given by the equation $x+y+z=10$ at A. Produce PA to B such that PA $=$ AB. Let C be the image of B in π. Let D be a point on the plane π such that $ABDC$ is a parallelogram. Let L_1 be the projection of L on π.
Question:
Coordinates of C are -

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(A) $\left(\frac{-1}{3}, \frac{10}{3}, \frac{7}{3}\right)$

(B) $\left(\frac{-1}{3}, \frac{-10}{3}, \frac{5}{3}\right)$

(C) $\left(\frac{1}{3}, \frac{10}{3}, \frac{7}{3}\right)$

(D) $\left(\frac{1}{3}, \frac{-10}{3}, \frac{-7}{3}\right)$

Q.14

Passage:

Let L be the line whose equations are $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{1}$. Let P represent the point $(1, 2, 3)$.L meets the plane π given by the equation $x + y + z = 10$ at A. Produce PA to B such that PA = AB. Let C be the image of B in π. Let D be a point on the plane π such that $ABDC$ is a parallelogram. Let L_1 be the projection of L on π.

Question:

Equation of L_1 is -

(A) $\frac{7-3x}{1} = \frac{3y-10}{2} = \frac{13-3z}{1}$

(B) $\frac{7-3x}{2} = \frac{3y-10}{2} = \frac{13-3z}{1}$

(C) $\frac{7-3x}{0} = \frac{3y-10}{1} = \frac{13-3z}{2}$

(D) $\frac{7-3x}{1} = \frac{3y-10}{2} = \frac{13-3z}{2}$

Q.15

Passage:

The lines $\mathrm{L}_1 : \frac{\mathrm{x}-4}{1} = \frac{\mathrm{y}+3}{-4} = \frac{\mathrm{z}+1}{7}$ and $\mathrm{L}_2 : \frac{\mathrm{x}-1}{2} = \frac{\mathrm{y}+1}{-3} = \frac{\mathrm{z}+10}{\mathrm{k}}$ are coplanar $-$

Question:

The value of k is -

(A) -8

(B) 4

(C) 8

(D) 16

Q.16

Passage:

The lines $\mathrm{L}_1 : \frac{\mathrm{x}-4}{1} = \frac{\mathrm{y}+3}{-4} = \frac{\mathrm{z}+1}{7}$ and $\mathrm{L}_2 : \frac{\mathrm{x}-1}{2} = \frac{\mathrm{y}+1}{-3} = \frac{\mathrm{z}+10}{\mathrm{k}}$ are coplanar $-$

Question:

Point of intersection of L_1 and L_2 is at -

(A) $(5, 7, 6)$

(B) $5, -7, 6)$

(C) $(5, 7, -6)$

(D) $(5, -7, -6)$

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Q.17

Passage:

The lines $L_1 : \frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$ and $L_2 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{k}$ are coplanar –

Question:

Equation of the plane passing through the origin and the point $(1, 0, -1)$ and which is normal to the plane containing L_1 and L_2 is -

(A) $x - y + z = 0$

(B) $2x + y + 2z = 0$

(C) $x + y + z = 0$

(D) $3x - y + 3z = 0$

Q.18

Match the column -

Column I	**Column II**
(A) If the plane $ax - by + cz = bd$ contains the line $\frac{x-a}{a} = \frac{y-3d}{b} = \frac{z-c}{c}$, then $\frac{b}{d}$ is equal to	(p) 1/3
(B) The distance of the point $(1, -3, 5)$ from the plane $2x - 4y + 3z - 7 = 0$ measured parallel to $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$ is equal to	(q) 4
(C) If the straight lines $\frac{x-2}{3} = \frac{y-3}{4} = \frac{4-z}{k}$ and $\frac{x-1}{k} = \frac{y-3}{2} = \frac{z-5}{1}$ intersect, then k is equal to	(r) $\frac{1}{\sqrt{35}}$
(D) The distance of the point $(1, -2, 1)$ from the plane $x - 3y + 5z = 11$ is	(s) $\frac{5}{\sqrt{29}}$
	(t) $\frac{77}{13}$

(A) (A) - q, (B) - q, (C) - t, (D) - t

(B) (A) - q, (B) - t, (C) - p, (D) - r

(C) (A) -p, (B) - t, (C) - p, (D) - r

(D) (A) - r, (B) - s, (C) - r, (D) - q

Q.19

Find the total number of points, whose perpendicular distances from yz, zx and xy-planes are in A.P and whose distances from x, y and z axes are $\sqrt{13}, \sqrt{10}$ and $\sqrt{5}$ respectively AP is increasing in nature

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Q.20

A line L_1 with direction ratios $(-3, 2, 4)$ passes through the point $A(7, 6, 2)$ and a line L_2 with direction ratios $(2, 1, 3)$ passes through the point $B(5, 3, 4)$. A line L_3 with direction ratios $(2, -2, -1)$ intersects L_1 and L_2 at C and D. Then find the value of CD.

Answers & Solutions

Q.1 Answer:

$\sqrt{2/7}$

Solution:

Equation of plane containing L_1 is given by,

$a(x-2)+b(y-1)+c(z+1)=0$

Now this plane is perpendicular to both the given lines

$\Rightarrow a+2c=0..(1)$

and $a+b-c=0..(2)$

Solving (1) and (2), we get $a=-2c$ **and** $b=3c$

Therefore plane is $-2(x-2)+3(y-1)+(z+1)=0$

$\Rightarrow -2x+3y+z+2=0$

Hence, distance of this plane from origin is

$=\left|\frac{2}{2^2+3^2+1^2}\right|=\sqrt{\frac{1}{7}}$

Q.2 Answer:

23

Solution:

Equation of the line is

$\frac{x-2}{1}=\frac{y+2}{-3}=\frac{z-5}{2}=\lambda$ **...(1)**

hence any point on the line (1) can be taken as

$$x=\lambda+2$$
$$\Rightarrow\ y=-(3\lambda+2)$$
$$z=(2\lambda+5)$$

for some λ point lies on the plane

$2x-3y+4z=163 \quad \ldots(2)$

$2(\lambda+2)+3(3\lambda+2)+4(2\lambda+5)=163$

This gives $19\lambda=133$

$\Rightarrow \lambda=7$

Hence, $P\equiv(9,-23,19)$

Also (1) intersect YZ plane i.e. $x=0$

$\Rightarrow \lambda+2=0,$ hence $\lambda=-2$

$\therefore Q(0,4,1)$

So, $PQ=\sqrt{9^2+27^2+18^2}=9\sqrt{1+3^2+2^2}=9\sqrt{14}$

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$\therefore \quad a = 9$ and $b = 14$

Hence, $a + b = 23$

Q.3 Answer:

$\overrightarrow{\mathrm{r}} = (-3\hat{\mathrm{i}} + 5\hat{\mathrm{j}} - 6\hat{\mathrm{k}}) + \lambda(2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} + 2\hat{\mathrm{k}})$

Solution:

Comparing the given equation with the standard form $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

We observe that $x_1 = -3, y_1 = 5, z_1 = -6; a = 2, b = 4, c = 2.$

Thus, the required line passes through the point $(-3, 5, -6)$ **and is parallel to the vector** $2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} + 2\hat{\mathrm{k}}$.

Let $\vec{r}$ **be the position vector of any point on the line, then the vector equation of the line is given by** $\overrightarrow{\mathrm{r}} = (-3\hat{\mathrm{i}} + 5\hat{\mathrm{j}} - 6\hat{\mathrm{k}}) + \lambda(2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} + 2\hat{\mathrm{k}})$

Q.4 Answer:

$\frac{9}{\sqrt{41}}$

Solution:

We have Line: $\frac{x-2}{1} = \frac{y-1}{1} = \frac{z+\frac{1}{2}}{-1/2} = t$ **(let)**

Plane: $x + 2y + 6z - 10 = 0$

Vector along the line, $\vec{V} = \hat{i} + \hat{j} - \frac{1}{2}\hat{k}$

Vector normal to the plane $\vec{n} = \hat{i} + 2\hat{j} + 6\hat{k}$

As $\overrightarrow{\mathrm{V}} \cdot \overrightarrow{\mathrm{n}} = 1 + 2 - \frac{1}{2} \cdot 6 = 0$

Hence line is parallel to the plane.

Now distance from the point $\left(2, 1, -\frac{1}{2}\right)$ **to the plane** $x + 2y + 6z - 10 = 0$ **is,**

$\mathrm{d} = \frac{\left|2+2-6\left(\frac{1}{2}\right)-10\right|}{\sqrt{36+4+1}} = \frac{9}{\sqrt{41}}$

Q.5 Answer:

$aa' + cc' + 1 = 0$

Solution:

Equation of the line $x = ay + b, z = cy + d$ **can be written as** $\frac{x-b}{a} = y, \frac{z-d}{c} = y \Rightarrow \frac{x-b}{a} = \frac{y-0}{1} = \frac{z-d}{c} \ldots\ldots(1)$

Also, $x = a'y + b', \quad z = c'y + d'$ **can be written as** $\frac{\mathrm{x-b'}}{\mathrm{a'}} = \mathrm{y}, \frac{\mathrm{z-d'}}{\mathrm{c'}} = \mathrm{y} \Rightarrow \frac{\mathrm{x-b'}}{\mathrm{a'}} = \frac{\mathrm{y-0}}{1} = \frac{\mathrm{z-d'}}{\mathrm{c'}} \ldots.(2)$

Lines (1) and (2) will be perpendicular to each other if $\ell_1\ell_2 + m_1m_2 + n_1n_2 = 0$

$\Rightarrow \quad aa' + 1 \times 1 + c \times c' = 0 \Rightarrow aa' + cc' + 1 = 0$

Q.6 Answer:

$\sin^{-1}\left(\frac{8}{21}\right)$

Solution:

Let θ **be the angle between the line and the normal to the plane.**

Converting the given equations into vector form, we have

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$\overrightarrow{\mathrm{r}} = (-\hat{\mathrm{i}} + 3\hat{\mathrm{k}}) + \lambda(2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 6\hat{\mathrm{k}})$ **and** $\overrightarrow{\mathrm{r}} \cdot (10\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 11\hat{\mathrm{k}}) = 3$

Here, $\overrightarrow{\mathrm{b}} = 2\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 6\hat{\mathrm{k}}$ **and** $\overrightarrow{\mathrm{n}} = 10\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 11\hat{\mathrm{k}}$

$\sin\phi = \left|\frac{(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+6\hat{\mathrm{k}})\cdot(10\hat{\mathrm{i}}+2\hat{\mathrm{j}}-11\hat{\mathrm{k}})}{\sqrt{2^2+3^2+6^2}\sqrt{10^2+2^2+11^2}}\right| = \left|\frac{-40}{7\times15}\right| = \left|\frac{-8}{21}\right| = \frac{8}{21}$

or $\phi = \sin^{-1}\left(\frac{8}{21}\right)$

Q.7 Answer:

$\left(\frac{4}{3}, \frac{-2}{3}, \frac{-2}{3}\right)$

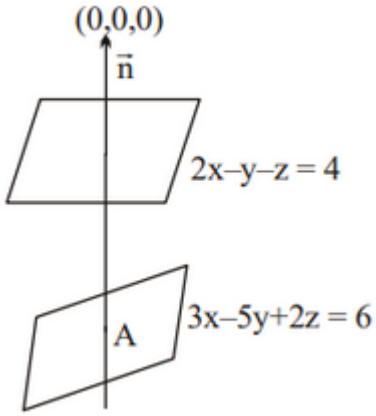
Solution:

Vector perpendicular to $2x - y - z = 4$ **is** $\vec{n} = 2\hat{i} - \hat{j} - \hat{k}$

Also line is collinear with $\vec{n}$.

So equation of line is $\vec{r} = \lambda(2\hat{i} - \hat{j} - \hat{k})$ **...(1)**

Now line (1) meet the plane $\vec{r} \cdot (3\hat{i} - 5\hat{j} + 2\hat{k}) = 6$



So, $\lambda(6 + 5 - 2) = 6$

$\Rightarrow \lambda = \frac{2}{3}$

Hence $p.v.$ **of the point '** A **' is** $\frac{2}{3}(2\hat{i} - \hat{j} - \hat{k})$

Q.8 Answer:

$\frac{5}{6}$

Solution:

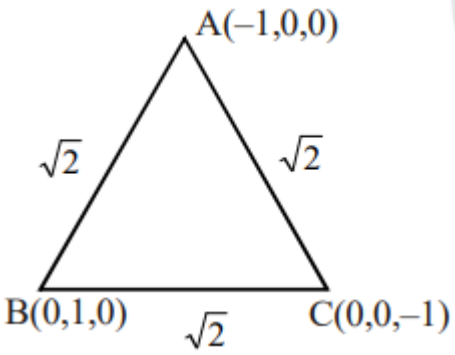
Clearly $\triangle ABC$ **is equilateral.**

$$\Rightarrow \frac{\mathrm{r}}{\mathrm{R}} = 4 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow 2\mathrm{r} = \mathrm{R}$$

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Also, $R = \frac{abc}{4\Delta} = \frac{2\sqrt{2}}{4 \cdot \frac{\sqrt{3}}{4} \cdot 2} = \sqrt{\frac{2}{3}} \Rightarrow r = \frac{1}{\sqrt{6}}$



$\therefore \quad R^2 + r^2 = \frac{1}{6} + \frac{2}{3} = \frac{5}{6}$

Q.9 Answer:

AB meets the plane at $(0, 4, 1)$

Solution:

Direction ratios of AB **are** $2, -3, 2$

$2 \times 2 - 3 \times 1 - 2 \times 1 \neq 0$

$\therefore$ AB **is not parallel to the plane.**

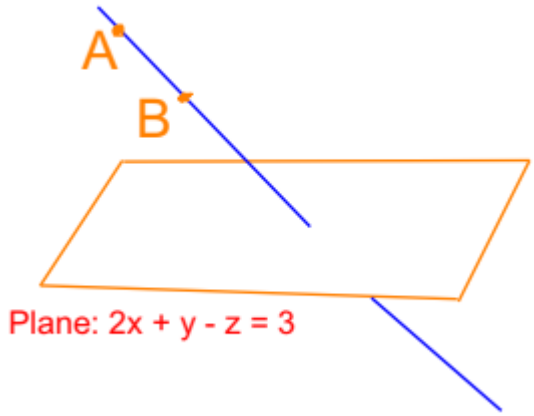
If AB **is divided by the plane in the ratio** $\lambda : 1$**, then the point of division is** $\left(\frac{4\lambda+2}{\lambda+1}, \frac{-2\lambda+1}{\lambda+1}, \frac{5\lambda+3}{\lambda+1}\right)$ **and this lies on the plane**

$\therefore \quad 2(4\lambda + 2) + (1 - 2\lambda) - (5\lambda + 3) = 3(\lambda + 1)$

This gives $\lambda = -\frac{1}{2}$

$\therefore (C)$ **is correct.**

As the plane divides AB **externally in the ratio** $1 : 2$ **so both points** A **and** B **lie on the same side of the plane.**



Q.10 Answer:

If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar and $\vec{d}$ is any vector, then $[\vec{d}\vec{b}\vec{c}]\vec{a} + [\vec{d}\vec{c}\vec{a}]\vec{b} + [\vec{d}\vec{a}\vec{b}]\vec{c} - [\vec{a}\vec{b}\vec{c}]\vec{d} = \vec{0}$

Solution:

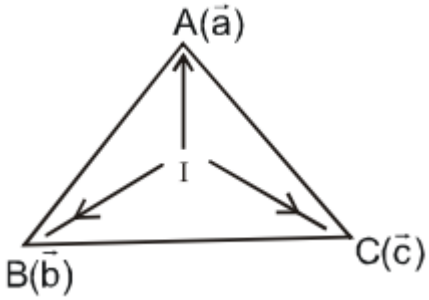
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(A) $\left[\begin{array}{lll}\vec{d} & \vec{b} & \vec{c}\end{array}\right]\vec{a}+\left[\begin{array}{lll}\vec{d} & \vec{c} & \vec{a}\end{array}\right]\vec{b}+\left[\begin{array}{lll}\vec{d} & \vec{a} & \vec{b}\end{array}\right]\vec{c}-[\vec{a}\vec{b} \quad \vec{c}]\vec{d}$

$=([\vec{b}\vec{c}\vec{d}]\vec{a}-[\vec{b}\vec{c}\vec{a}]\vec{d})+([\vec{d}\vec{a}\vec{b}]\vec{c}-[\vec{d}\vec{a}\vec{c}]\vec{b})=(\vec{b}\times\vec{c})\times(\vec{a}\times\vec{d})+(\vec{d}\times\vec{a})\times(\vec{c}\times\vec{b})$

$=(\vec{d}\times\vec{a})\times(\vec{b}\times\vec{c})-(\vec{d}\times\vec{a})\times(\vec{b}\times\vec{c})=\overrightarrow{0}$

(B)

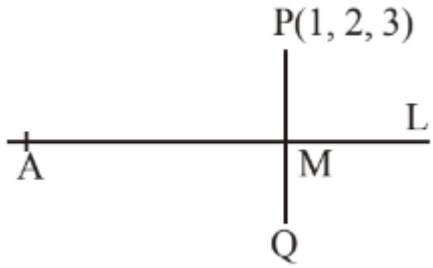


$I\equiv\frac{a\vec{a}+b\vec{b}+c\vec{c}}{a+b+c}$

Q.11 Answer:

15

Solution:



Let L **be the line** $\frac{x}{-2}=\frac{y+1}{2}=\frac{z-1}{1}$

A point A **on the line is** $\mathrm{A}(0,-1,1)$.

D. Rs of L **are** $-2,2,1$

D.Cs are $\frac{-2}{3},\frac{2}{3},\frac{1}{3}$

Let P **be** $(1,2,3)$

Let $Q\left(x_1,y_1,z_1\right)$ **be the image of** P.

PQ **meets** L **at** $\mathrm{M}.\mathrm{PM}=\mathrm{MQ}$

Equation of L **is** $\frac{x}{\frac{-2}{3}}=\frac{y+1}{\frac{2}{3}}=\frac{z-1}{\frac{1}{3}}=r$

Let $\mathrm{AM}=\mathrm{r}$

M **is given by** $\left(-\frac{2\mathrm{r}}{3},-1+\frac{2\mathrm{r}}{3},1+\frac{\mathrm{r}}{3}\right)$

P **is** $(1,2,3)$

D. Rs of PM are $\left(1+\frac{2\mathrm{r}}{3},3-\frac{2\mathrm{r}}{3},2-\frac{\mathrm{r}}{3}\right)$

PM and L are perpendicular

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$\therefore \quad -2\left(1+\frac{2r}{3}\right)+2\left(3-\frac{2r}{3}\right)+1\left(2-\frac{r}{3}\right)=0; 6-\frac{9r}{3}=0; r=2$

$\therefore \quad$ M is $\left(-\frac{4}{3}, \frac{1}{3}, \frac{5}{3}\right)$

$P(1,2,2) \& Q\left(x_1, y_1, z_1\right)$; Midpoint of PQ is $\left(\frac{x_1+1}{2}, \frac{y_1+2}{2}, \frac{z_1+3}{2}\right)$

This is the same as $M\left(-\frac{4}{3}, \frac{1}{3}, \frac{5}{3}\right)$

$\therefore \quad \frac{x_1+1}{2}=-\frac{4}{3} \Rightarrow 3x_1+3=-8 \Rightarrow x_1=\frac{-11}{3}$

$\frac{y_1+2}{2}=\frac{1}{3} \Rightarrow 3y_1+6=2 \Rightarrow y_1=-\frac{4}{3}$

$\frac{z_1+3}{2}=\frac{5}{3} \Rightarrow z_1=\frac{1}{3}$

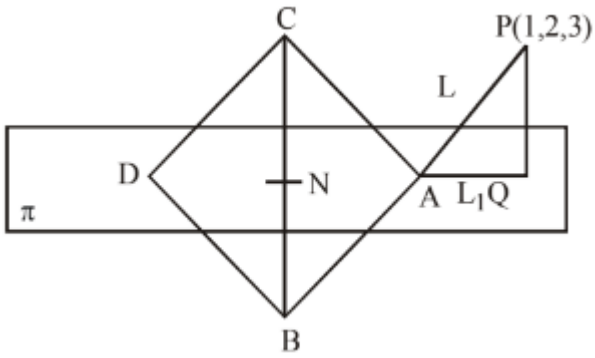
$\therefore \quad Q$ **is** $\left(-\frac{11}{3}, \frac{-4}{3}, \frac{1}{3}\right)$

$OQ^2=\frac{121}{9}+\frac{16}{9}+\frac{1}{9}=\frac{138}{9}=\frac{46}{3}$

Q.12 Answer:

$(3,6,5)$

Solution:



(i) Let L **meet the plane** π **at A. Let** Q **be the foot of the perpendicular from** P **on** π**. AQ is denoted by** L_1**.** PA **is produced to** B **such that** $\mathrm{PA}=\mathrm{AB}$**.** C **is the image of** B **in** π**. D lies on** π **and** ABDC **is a parallelogram.**

Any point on L **is** $(1+\mathrm{r}, 2+2\mathrm{r}, 3+\mathrm{r})$

If this lies on the plane $x+y+z=10$ **we have** $1+r+2+2r+3+r=10$

$4\mathrm{r}+6=10$

$4r=4$

$r=1$

Point A **is** $(2,4,4)$

A is the mid point of PB

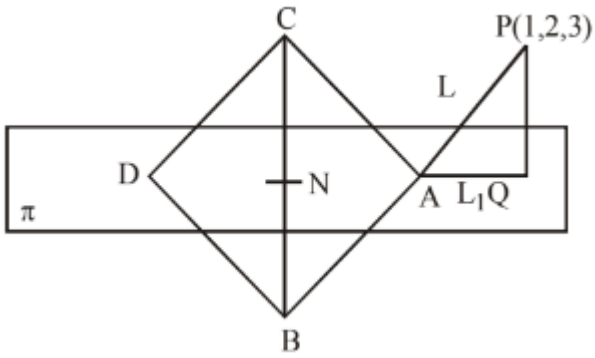
$\therefore \mathrm{B}$ **is** $(3,6,5)$

Q.13 Answer:

$\left(\frac{1}{3}, \frac{10}{3}, \frac{7}{3}\right)$

Solution:

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(ii) BC is $\frac{x-3}{1}=\frac{y-6}{1}=\frac{z-5}{1}=R$.

$(R+3, R+6, R+5)$ **is a point on the line** BC **It it is** N, $\quad 3R+14=10$

$3R=-4, R=-4/3$

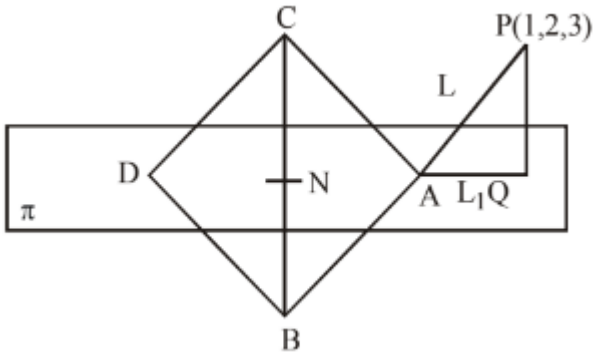
$\Rightarrow \quad R$ **corresponding to the image** C **is** $\frac{-8}{3}$

Hence, C **is** $\left(\frac{-8}{3}+3, \frac{-8}{3}+6, \frac{-8}{3}+5\right)$ **i.e.** $\left(\frac{1}{3}, \frac{10}{3}, \frac{7}{3}\right)$

Q.14 Answer:

$\frac{7-3x}{1}=\frac{3y-10}{2}=\frac{13-3z}{1}$

Solution:



(iii) Line through P normal to π **is** $\frac{x-1}{1}=\frac{y-2}{1}=\frac{z-3}{1}=r; r+1+r+2+r+3=10; 3r=4, r=\frac{4}{3}$

Q **is** $\left(\frac{7}{3}, \frac{10}{3}, \frac{13}{3}\right)$, A **is** $(2,4,4)$

Therefore, L_1 **is** $\frac{x-\frac{7}{3}}{-\frac{1}{3}}=\frac{y-\frac{10}{3}}{\frac{2}{3}}=\frac{z-\frac{13}{3}}{-\frac{1}{3}} \Rightarrow \frac{3x-7}{-1}=\frac{3y-10}{2}=\frac{3z-13}{-1}$

Q.15 Answer:

8

Solution:

(i) Any plane through L_1 **is of the form**

$a(x-4)+b(y+3)+c(z+1)=0$ **.....(1)**

where $a-4b+7c=0$ **.....(2)**

If L_1 **and** L_2 **are coplanar, (1) should contain** L_2

$2a-3b+kc=0$ **.....(3)**

$-3a+2b-9c=0$ **.....(4)**

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Eliminating a, b, c from $(2), (3)$ and (4) we have

$$\begin{vmatrix} 1 & -4 & 7 \\ 2 & -3 & k \\ -3 & 2 & -9 \end{vmatrix} = 0 \Rightarrow -10k + 80 = 0 \Rightarrow k = 8$$

Q.16 Answer:

$5, -7, 6)$

Solution:

(i) Any plane through L_1 is of the form

$a(x-4) + b(y+3) + c(z+1) = 0$ **.....(1)**

where $a - 4\,b + 7c = 0$ **.....(2)**

If L_1 and L_2 are coplanar, (1) should contain L_2

$2a - 3\,b + kc = 0$ **.....(3)**

$-3a + 2\,b - 9c = 0$ **.....(4)**

Eliminating a, b, c from $(2), (3)$ and (4) we have

$$\begin{vmatrix} 1 & -4 & 7 \\ 2 & -3 & k \\ -3 & 2 & -9 \end{vmatrix} = 0 \Rightarrow -10k + 80 = 0 \Rightarrow k = 8$$

(ii) Any point on L_1 is $P(4 + r, -3 - 4r, -1 + 7r)$

Any point on L_2 is $Q(1 + 2\,s, -1 - 3\,s, -10 + 8\,s)$

L_1 and L_2 are coplanar and they are not parallel.

$\Rightarrow$ L_1 and L_2 intersect

$\Rightarrow$ For same value of r and s we must have

$4 + r = 1 + 2\,s; -3 - 4r = -1 - 3\,s; -1 + 7r = -10 + 8\,s$

(i.e.) we have

$r - 2s = -3 \ldots\ldots(1)$

$4r - 3s = 2 \ldots\ldots(2)$

$7r - 8s = -9 \ldots\ldots(3)$

Solving (1), (2) & (3)

Solving (1) and (2), we get $r = 1, s = 2$. These values satisfy (3) also

Substituting $r = 1$ or $s = 2$ we get the point of intersection as $(5, -7, 6)$

(iii) D.R of normal to the plane containing L_1 and L_2

$$\left.\begin{matrix} 1 & -4 & 7 \\ 2 & -3 & 8 \end{matrix}\right\} \Rightarrow [-32 + 21, 6, 5] \text{ or } [-11, 6, 5]$$

Let the plane be $Ax + By + Cz = 0$ (constant term $= 0$, since plane passes through the origin)

We have $A - C = 0$ (plane passes through $(1, 0, -1)$)

And $-11\,A + 6\,B + 5C = 0 \Rightarrow B = C = A$

Plane is $x + y + z = 0$

Q.17 Answer:

$x + y + z = 0$

Solution:

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(iii) D.R of normal to the plane containing L_1 and L_2

$$\left.\begin{matrix} 1 & -4 & 7 \\ 2 & -3 & 8 \end{matrix}\right\} \Rightarrow [-32+21, 6, 5] \text{ or } [-11, 6, 5]$$

Let the plane be $Ax + By + Cz = 0$ (constant term $= 0$, since plane passes through the origin)

We have $A - C = 0$ (plane passes through $(1, 0, -1)$)

And $-11\ A + 6\ B + 5C = 0 \Rightarrow B = C = A$

Plane is $x + y + z = 0$

Q.18 Answer:

(A) - q, (B) - t, (C) - p, (D) - r

Solution:

$(A) - q, (B) - t, (C) - p, (D) - \mathbf{r}$

(A) We have $a^2 - b^2 + c^2 = 0$ and $a^2 - 3bd + c^2 = bd$

$\Rightarrow \quad b^2 = 4bd$, **as** $b \neq 0$

$\Rightarrow \quad \frac{b}{d} = 4$

(B) Equation of line through $(1, -3, 5)$ and parallel to given line is $\frac{x-1}{2} = \frac{y+3}{3} = \frac{z-5}{-6}$

Any point on it $2r + 1, 3r - 3, -6r + 5$

This point lies on the plane if $r = \frac{11}{13}$

So, the point is $\left(\frac{35}{13}, \frac{-6}{13}, \frac{-1}{13}\right)$

Desired distance $= \sqrt{\left(\frac{35}{13} - 1\right)^2 + \left(\frac{-6}{13} + 3\right)^2 + \left(\frac{-1}{13} - 5\right)^2} = \frac{77}{13}$

(C) General points on two given lines may be written as $(3r_1 + 2, 4r_1 + 3, -kr_1 + 4)$ and $(kr_2 + 1, 2r_2 + 3, r_2 + 5)$

If two lines intersect, then for some r_1 and r_2,

$3r_1 + 2 = kr_2 + 1, 4r_1 + 3 = 2r_2 + 3, -kr_1 + 4 = r_2 + 5.$

Eliminating r_1 and r_2 we get, $-6k + 2 = 0 \Rightarrow k = 1/3$

(D) Required distance $= \frac{1-3(-2)+5(1)-11}{\sqrt{1^2+3^2+5^2}} = \frac{1}{\sqrt{35}}$

Q.19 Answer:

—

Solution:

As perpendicular distances are in A.P.

Let $|a - d|, |a|, |a + d|$ be these distances.

So, the coordinates of point P are $(\pm(a - d), \pm a, \pm(a + d))$

Now, distances from P of x-axis, y-axis and z-axis are respectively,

$2a^2 + d^2 - 2ad = 5$ **.....(1)**

$2a^2 + 2\ d^2 = 10$ **.....(2)**

$2a^2 + d^2 + 2ad = 13$ **.....(3)**

Solving eq. (1) and (3), we get, $4ad = 8$

Substituting in (2), $\frac{4}{d^2} + d^2 = 5 \Rightarrow d^2 = 1$ **or 4**

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$\Rightarrow \quad d = \pm 1, a = \pm 2$

So, points are $(\pm 1, \pm 2, \pm 3)$. (There will be 8 such points)

Q.20 Answer:

—

Solution:

Equation of L_1 are $\frac{x-7}{-3} = \frac{y-6}{2} = \frac{z-2}{4} = r_1$ so that any point on it is $[7 - 3r_1, 6 + 2r_1, 2 + 4r_1]$.
Let this point be C. Equations of line L_2 are $\frac{x-5}{2} = \frac{y-3}{1} = \frac{z-4}{3} = r_2$ so that any point on it is $[5 + 2r_2, 3 + r_2, 4 + 3r_2]$.
Let this point be D.
Now, C and D lie on line L_3 with d.r. $(2, -2, -1)$.
Hence, $\frac{2-3r_1-2r_2}{2} = \frac{3+2r_1-r_2}{-2} = \frac{-2+4r_1-3r_2}{-1}$
or, $r_1 + 3r_2 = 5$ and $6r_1 - 5r_2 = 7 \Rightarrow r_1 = 2, r_2 = 1$
Hence, the points C and D are $(1, 10, 10)$ and $(7, 4, 7)$
$\Rightarrow \quad CD = \sqrt{36 + 36 + 9} = \sqrt{81} = 9$

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